

Aarón Ayllón Benítez, PhD

AI & Data Science Leader — Enterprise ML, GenAI & Agentic Platforms · Team and Portfolio Ownership

Madrid, Spain · +34 608 236 720 · ayllonbenitez.aaron@gmail.com · aaronayllonbenitez.es

EXECUTIVE SUMMARY

Data scientist turned AI/ML leader (PhD Computer Science, MSc Artificial Intelligence) who has spent five-plus years building and running multidisciplinary data science teams and a ~€1.3M platform budget inside a global corporation. Scaled a team from 8 to 16 data scientists, ML/software engineers and solution designers, and owned two production AI/ML platforms covering the full analytics journey from descriptive to predictive, delivering recurring savings above €3.5M. Owns the portfolio calls that follow — what to scale, what to hand over and what to shut down — alongside vendor strategy, demand intake and prioritisation across departments and European markets. Combines hands-on technical depth in ML, LLMs and agentic workflows, still shipping production code solo as co-founder of a live B2B SaaS platform, with life-sciences training in biochemistry, bioinformatics and biomedical machine learning on patient-derived data, and a track record of translating analytics into quantified business cases defended in front of senior leadership.

€3.5M+

recurring savings from
portfolio decisions

8 → 16

team scaled, hired and developed

~90%

less effort per new
project after standardisation

~100

the whole of BASF's global
data science community

CORE SKILLS

AI & Data Science: Machine Learning, Deep Learning, Computer Vision, Time-series Forecasting, MLOps, Agentic workflows (multi-step reasoning, tool calling, evaluation), Descriptive → Predictive Analytics, GenAI & LLMs, Knowledge Graphs & Ontologies

Leadership: Hiring & team building, Mentoring managers and junior scientists, Cross-functional alignment, Multi-market coordination, Performance and underperformance management, Career paths, Executive communication & storytelling

Portfolio & Delivery: Portfolio governance & rationalisation, Budget ownership & P&L, Vendor evaluation & build-vs-buy decisions, Product line management, Build / scale / hand-over / sunset decisions, Product strategy & roadmap, Demand intake & prioritisation, Enterprise platform delivery

Technical: Python, R, SQL, advanced statistics, Azure & Azure DevOps, APIs / MCP, data engineering workflows, CI/CD, Power BI, AI development tools

Domains: Life sciences, healthcare & biomedical data, pharma and agricultural R&D, industrial and commodity business

PROFESSIONAL EXPERIENCE

BASF Digital Solutions S.L. — Madrid, Spain

Head of Advanced Imaging & Forecasting Solutions — AI & ML Platforms

2022 – Present

Official BASF title: Product Manager · scope equivalent to Head of Product / Global Product Line Manager

Own a ~€1.3M annual platform budget and a 16-person team (scaled from 8: data scientists, ML/software engineers, solution designers) across BASF's computer vision, MLOps and forecasting platforms, serving the company's global data science community. Accountable for portfolio strategy, vendor decisions and demand prioritisation across three departments and multiple European markets, reporting monthly to Finance and presenting investment cases at VP level.

- Scaled the team from 8 to 16 through structured hiring and career-path design, directly mentoring both managers and junior scientists and owning performance — including underperformance — management
- Transferred the CV/MLOps platform to its best-fit owner (~€2M recurring savings, 70 reusable templates, ~90% effort reduction for ~100 data scientists) and closed the forecasting platform after ROI review (~€1.5M savings)
- Instituted peer-reviewed governance for the 70-template library and tracked platform health via active-user adoption and time saved per user, validated through internal satisfaction surveys
- Secured €560K in funding — GenAI-powered labelling automation plus five further innovation initiatives — presenting quantified business cases directly to VP stakeholders
- Solidified an exclusive delivery partnership with Tecnia, ran a proof-of-concept evaluation of SIGMA AI for image-labelling, and benchmarked competing tools to inform build-vs-buy decisions
- Co-led BASF's cross-departmental Phenotyping working group with Agriculture, standardising multi-site image acquisition and integrating the CV platform with adjacent phenotyping tools

Technical Product Manager — Ontology & Semantic Layer

2021 – 2022

Official BASF title: Product Component Manager · foundation for the AI & ML platform portfolio taken on next

Owned BASF's Ontology Management Framework (OMF) — enterprise strategy, governance and technology stack (Protégé, RDFlib, Virtuoso, automated curation pipeline) for bio-ontologies underpinning downstream AI and agentic systems, including ontologies authored in-house. Led 5 engineers and 2 external partners, evaluated Stardog as an alternative before retaining and extending the incumbent stack, and communicated ontology strategy and adoption from general-audience sessions up to VP-level stakeholders across the business.

- Delivered 20 bio-ontologies, several authored in-house, adopted as the vocabulary standard across the R&D Agricultural Solutions department by every internal tool within 6–12 months
- Co-authored BASF's GOMO ontology governance framework (published, ISMB 2022) with the Ontology Engineering Group at Universidad Politécnica de Madrid
- Led the EPPO Ontology development for plant and pest code representation, published in Frontiers in Artificial Intelligence and released as open source
- Established and led a biweekly data steward community of 10 governing vocabulary standardisation, processing 10–15 terminology requests per month

Umeå Plant Science Centre — Umeå, Sweden

Postdoctoral Researcher, Machine Learning

2020 – 2021

- Awarded the Kempestiftelserna Fellowship and two further competitive grants secured as main author (KSLA, SPPS) to pursue a focused research niche combining ontology-based semantic similarity methods from my PhD with deep learning to infer functional gene annotation in non-model organisms — e.g. *Picea abies* (Norway spruce) — where multi-decade growth cycles make direct experimental validation infeasible, inferring annotation via mapping against characterized model-organism genomes, a homology-based transfer strategy
- Designed production-oriented ML pipelines, containerized with Docker and run on the university's HPC cluster, with reproducibility, auto-scaling and experiment tracking — later used as the starting point for Master's students' thesis work
- Contributed to a doctoral student's research, co-authoring "[lincRNA annotation in Norway spruce](#)" (Physiologia Plantarum, 2024), while supporting other PhD students during the fellowship
- Served as Guest Editor for a special issue on AI in non-model plants in Plants (MDPI, Q1), managing peer review for ~10 submitted articles over the course of the fellowship
- Delivered 25 hours of Master's-level teaching and supported faculty in introductory bioinformatics courses

Université de Bordeaux / LaBRI — Bordeaux, France

PhD Researcher, Computer Science

2016 – 2019

Ministerial funding (MESR)

- Originated an ontology-based semantic-similarity methodology for gene-set annotation — the approach that would later underpin ontology governance work at BASF (GOMO, EPPO) — evaluating nine semantic similarity measures over Gene Ontology terms to identify the optimal trade-off between minimizing retained annotation terms and maximizing gene coverage, outperforming standard enrichment-analysis tools
- Built the method into GSA_n, a public gene-set annotation web platform still in active use today (gsan.labri.fr, 18+ citations), including as the research foundation for a subsequent PhD student under my supervisors
- Co-supervised a Master's student on parallel research benchmarking GSA_n against existing gene-set annotation tools
- Ran an applied-ML engagement funded by Bordeaux University Hospital during the closing months of the thesis, applying MHC epitope prediction to patient-derived HIV sequence data and surfacing a promising, unpublished finding — hands-on biomedical machine learning in a clinical research setting

INSERM — Bordeaux, France

Bioinformatics Engineer

2015 – 2016

- Ran a solo, fully autonomous pre-doctoral engagement integrating genomic and transcriptomic data from HIV patients using standard Gene Set Enrichment Analysis (GSEA), reporting directly to Rodolphe Thiébaut and my future PhD supervisors throughout the engagement
- Identified the limitations of standard enrichment analysis on real multi-omics data — the insight that directly seeded the semantic-similarity methodology later developed into GSA_n during my PhD — and published the resulting work as first author in PLoS ONE (2018), with Thiébaut as co-author

LEADERSHIP APPROACH

Build the bench, not just the backlog. Doubled the team through deliberate hiring, individual career paths and coaching of the managers who report into the role — including the hard conversations around underperformance rather than routing around them, because a team's ceiling is set by its weakest link.

Decide with evidence, and be willing to kill your own product. Recommended closing a platform I owned once the ROI and adoption data no longer supported it, and transferred another to a better-fit owner — both decisions defended with numbers in front of senior leadership.

Translate between worlds. Comfortable challenging model choices with ML engineers in the morning and framing the same problem as a business case for executives in the afternoon; this is where most analytics value is won or lost.

EDUCATION

MSc Artificial Intelligence — Universidad Internacional de La Rioja (UNIR) 2023–2024

PhD Computer Science — Université de Bordeaux / LaBRI, ministerial funding (MESR) 2016–2019

MSc Bioinformatics — University of Murcia 2013–2015

BSc Biochemistry — University of Murcia 2009–2013

RESEARCH & INDUSTRY VISIBILITY

Selected publications: Nature Communications (2021, contributor); NAR Genomics and Bioinformatics (Q1, 2020, first author); PLoS ONE (2018, first author); IEEE Information Visualization, London (2017, first author, invited talk); EPPO Ontology, Frontiers in Artificial Intelligence (2023, first author); GOMO ontology governance model (ISMB 2022). h-index 5; 132 citations.

Speaking & media: Industry showcase speaker at EXPOQUIMIA 2023 on advanced image analysis for the chemical industry, plus roundtable participation; podcast guest on AI in industry (Tecnología, Pasión y Futuro) and with the AI Cluster of the Community of Madrid.

Editorial & funding: Guest Editor, Plants (MDPI, Q1) special issue on gene annotation; Kempestiftelserna Fellowship and 2 competitive grants secured as main author (KSLA, SPPS).

Teaching & mentoring: 100+ hours of university teaching (Umeå, Bordeaux); mentored 25+ Master's and Bachelor's students across four academic years.

ENTREPRENEURSHIP

Proyecto Vigía — Spain

Co-founder (50%) & Sole Developer

2026 – Present

Co-founded and single-handedly designed and built, end to end, a B2B marketplace platform for a regulated Spanish services sector: Java/Spring Boot backend, React/TypeScript frontend, Keycloak-based identity and access management, geospatial PostgreSQL/PostGIS, document-processing pipeline with OCR, GDPR-compliant data handling, and Verifactu-compliant e-invoicing — from architecture through production deployment.

- Took the platform from architecture to closed beta solo in ~3 months (June–September 2026), using AI-assisted development tooling (Claude Code) to accelerate delivery, and is now collecting structured feedback from the closed beta cohort ahead of a planned public launch
- Conducted a self-led security review of the platform ahead of beta launch, identifying and remediating several vulnerabilities before launch
- Optimised infrastructure cost by architecting on ARM-based cloud instances (Hetzner), balancing performance against a bootstrapped budget without compromising reliability
- Evaluating monetisation models — transaction-based commission versus a premium subscription tier — to shape the platform's go-to-market approach

LANGUAGES

Spanish — native. English — full professional proficiency. French — full professional proficiency.